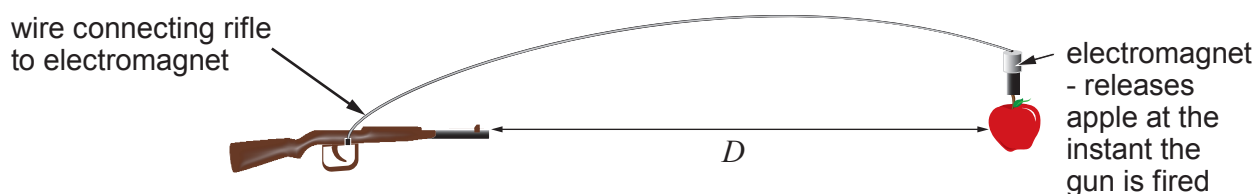
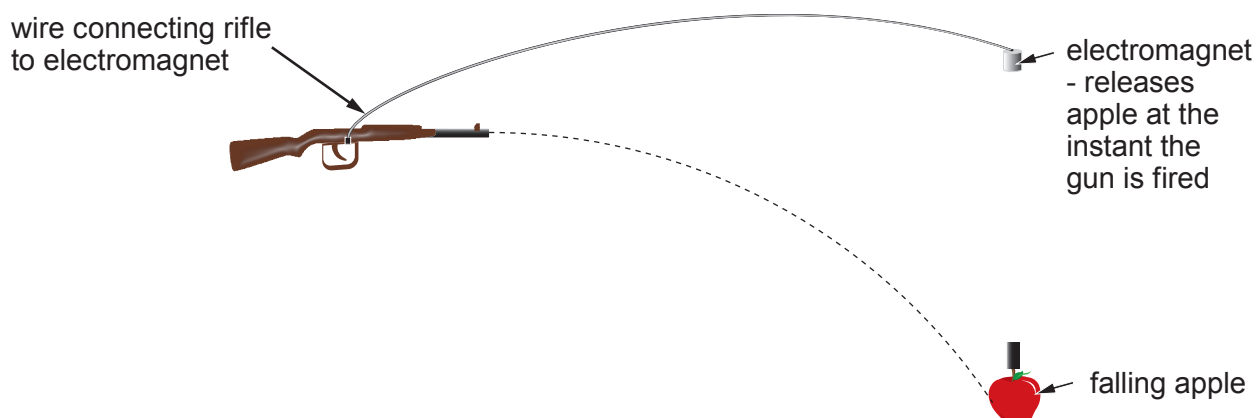


3. (a) The following demonstration is sometimes used to explain projectile motion. In the version shown, a toy rifle is situated a horizontal distance, D , from an apple suspended at a large height above the ground.

Initial set-up



Initially, the rifle is aimed horizontally at the apple. **At the instant** the gun is fired an electromagnet releases the apple so that it falls vertically. The path of the pellet from the gun is shown below.



- (i) Describe and explain the motion of the pellet from the instant the gun is fired. [Ignore the effects of air resistance.] [2]

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- (ii) The distance D is now increased and the experiment repeated. Describe one difference **and** one similarity that an observer viewing the apple would see in this case. [2]

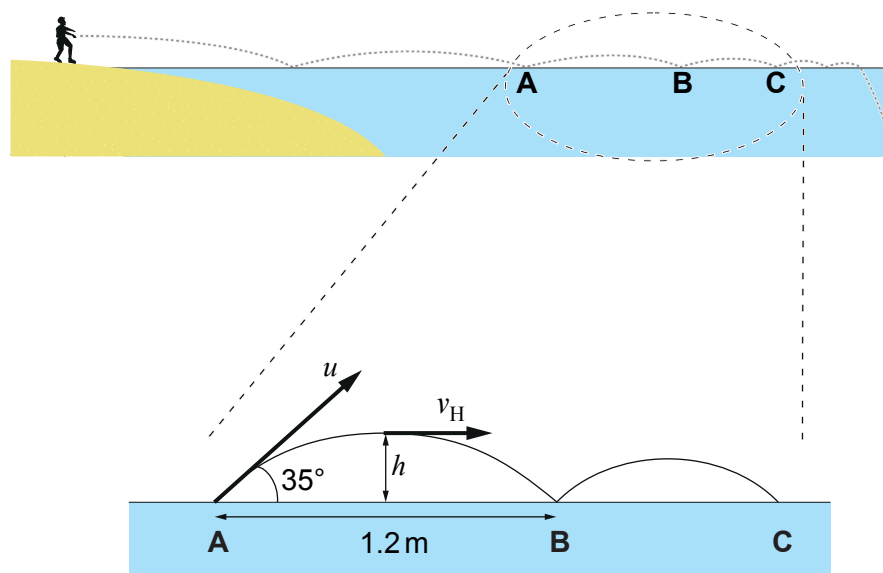
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- (b) The diagram shows a person 'skimming' a stone across the surface of a still pond. A magnified view of part of the stone's motion is also shown. [Ignore the effects of air resistance.]



- (i) Calculate v_H , the horizontal component of the velocity, u , given that the time taken for the stone to move from **A** to **B** is 0.40 s. [1]

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- (ii) Calculate h , the maximum height achieved by the stone between **A** and **B**. [4]

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- (iii) Calculate the **total** energy of the stone at height, h , given that it has a mass of 0.10 kg. [3]

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- (c) The thrower believes that the stone loses 20 % of its energy every time it impacts with the water. Investigate whether or not this is true **for the impact at B**, given the fact that the stone's velocity just before impacting the water **at C** is 3.3 m s^{-1} . [3]

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